

TILE SELECT PANEL

OVERVIEW

The Tile Select Panel is the panel on the right side of the screen that displays the buttons for all the different tiles. Please note that the system that manages the tool tips for the tiles is a separate system covered by the doc [Tool_Tip_System.pdf](#). The Tile Select Panel is primarily managed by the script `TileSelectPanel.cs`. It also has an overly co-dependent relationship with the script `TileSelectPanelScroll.cs`. The `TileSelectPanel` class primarily manages making buttons appear and tracking when they're pressed. I use the `TileSelectPanelScroll` class to allow the player to scroll through the buttons because Unity's built in scrolling system made me wonder if it was too late to switch to a sports science major. But for this doc, I'll mostly focus on the `TileSelectPanel` class.

HOW THE TILE SELECT PANEL WORKS

The tile select panel has three major responsibilities, which I'll go through one at a time. 1) Enabling new buttons. 2) Disabling buttons if there's not enough money, too much carbon, etc. 3) Telling the building system when the player selects a tile.

The Progression Manager tells the tile select panel when it should unlock new buttons, using the `AddButton()` function. When that happens, the tile select panel turns on the button's graphic and then rearranges/spaces all of the buttons to make room for it. It rearranges the buttons using the `updateButtonPositions` function, which goes through each enabled element in the panel and correctly spaces it.

The Tile Select Panel can disable buttons & turn them red if they can no longer be placed. To do this, at the beginning of the game, it adds listeners for all of the events that could cause a button to be un-activatable, including after the money & carbon ticks, whenever the amount of money is changed, a tile is placed, and whenever the player switches the current area. Whenever any of those events occur, it will call the `CheckPlaceabilityOfTiles` function, which goes through each button's tile, calling its `CheckIfTileIsPlaceable` function. If any of the tiles return "false", for unplaceable, the panel will disable that button, turning it red. Any tiles that return "true" for placeable will be enabled.

The Tile Select Panel's final responsibility is telling the building system (`BuildingSystem.cs`) when a tile is placed. It does this with the `clickButton` function. Every time a button is clicked, it passes its own reference to the `TileSelectPanel` class. If the button is enabled, the tile select panel will call the Building System's function `InitializeWithObject`, passing the tile prefab of the tile to place.

INTERNAL API

The TileSelectPanel has a singleton accessible at TileSelectPanel.TSP.

These are some of the most important public functions of the TileSelectPanel class:

void clickButton(GameObject clickedButton)

Checks if the tile of the clicked button is currently placeable and then tells the building manager that the tile has been selected if it is in fact placeable. This is called by buttons when they're clicked.

GameObject GetLowestActiveButton()

Returns the game object of the button that is currently lowest on the tile select panel. This is useful when the "NEW TILE!!!" pop up appears and the user clicks on it to scroll to the newest tile at the bottom.

void AddButton(buttonScript _buttonScript)

Enables & displays the button that corresponds to the input "_buttonScript"

void RemoveButton(GameObject prefabOfButton)

Hides the button that corresponds to the input "prefabOfButton".

void deselectAllButtons()

If any tile is selected, it deselects it.

float GetHeightOfButtons()

Returns the total pixel height of all the buttons combined.

bool isMouseOverScrollingPanel()

Returns true if the tile is over the tile select panel; returns false otherwise.

void updateButtonPositions()

Repositions all the activated buttons. This is useful when new buttons are added to ensure that they are all properly spaced.

-Graydon Bruyn, May 2026